



PRACTICAL WORK "ASTRONOMICAL DATA PROCESSING"
MASTER 2 COMPU**PHYS**

Photometric analysis of a Kuiper-Belt Object

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1 Introduction

This report presents the results of a photometric analysis of Kuiper Belt Object 29981 (1999 TD10), using images from the La Silla Observatory, captured on October 8th and 9th, 2001. The objective was to derive the object's rotation period and study its luminosity variations by constructing a light curve.

The analysis was performed in two stages:

- Computing magnitudes and positions and an animation of the movement of the object, using `'photometric.py'`
- Visualizing the light curve and fitting a sinusoidal model to extract the rotation period, using `'plots.ipynb'`

2 Principles of Photometric Analysis

Photometric analysis measures an object's brightness and its temporal variations. For this study, the key steps were:

1. **Flux Measurement:** The light intensity (flux) from 1999 TD10 and a reference star was measured using aperture photometry.
2. **Instrumental Magnitudes:** Flux values were converted to instrumental magnitudes to quantify the observed brightness.
3. **Calibration:** Magnitudes were calibrated using Pan-STARRS data (`PanSTARRS_8_oct_r.fits` and `PanSTARRS_9_oct_r.fits`) to align with a standard photometric system.
4. **Time Stamps:** The Modified Julian Day (MJD), extracted from FITS headers, was used to track temporal variations.
5. **Centroiding:** Accurate positions of the object and reference star were determined for reliable photometric measurements.

Coordinates of the Target and Reference Stars: The pixel coordinates of 1999 TD10 and the reference stars during observations were:

October 8th, 2001:

- Target: (1149, 1156)
- Reference Star: (1141, 1188)

October 9th, 2001:

- Target: (1169, 1168)
- Reference Star: (737, 1394)

3 Data Files

The following files were generated during the analysis:

- `8_OCT_mag_mjd.txt` and `9_OCT_mag_mjd.txt` contain the R magnitudes and MJDs for 1999 TD10.
- `8_OCT_pos_ref_obj.txt` and `9_OCT_pos_ref_obj.txt` provide the coordinates of the target and reference star for each observation.

4 Light Curve of 1999 TD10

The light curve, a plot of R magnitude versus MJD, reveals the periodic brightness variations of 1999 TD10. These variations are attributed to the object's rotation.

Figure 1 shows the light curve for 1999 TD10, with data points indicating measured magnitudes over time.

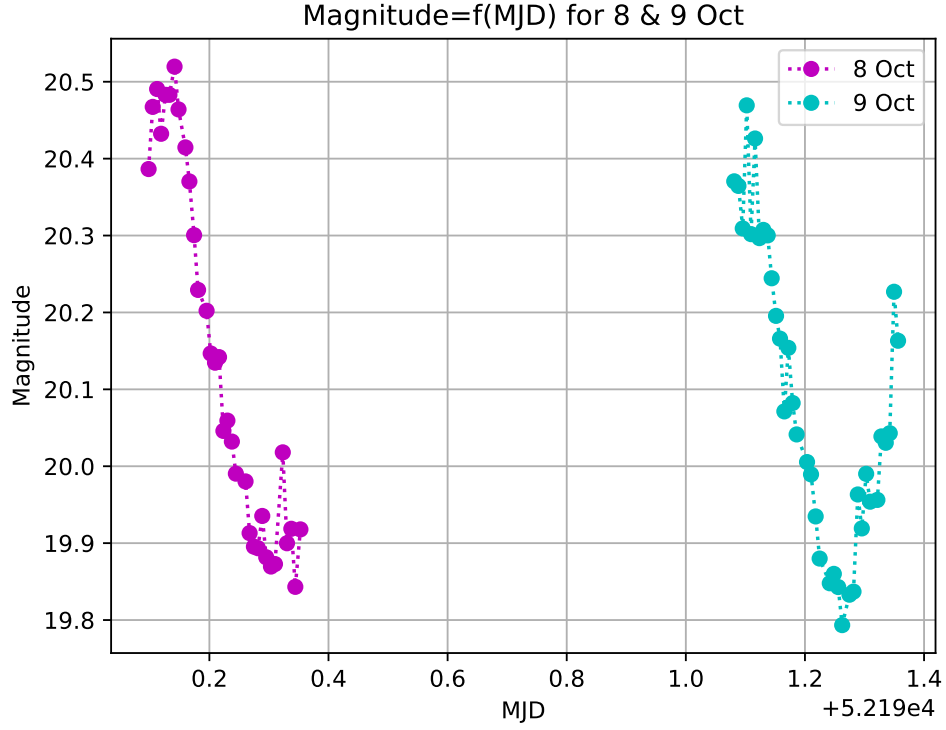


Figure 1: Light curve of 1999 TD10 showing brightness variations over time.

5 Rotation Period and Fitted Light Curve

The light curve was fitted with a sinusoidal model:

$$f(t) = a \sin(2\pi f(t + \phi)) + c$$

where a is the amplitude, f the frequency, ϕ the phase, and c the vertical shift.

The fit parameters were:

$$\begin{aligned} a &= 0.278, \\ f &= 3.100, \\ \phi &= 0.726, \\ c &= 20.125. \end{aligned}$$

The resulting rotation period, derived from the frequency, is approximately 15 hours, 29 minutes, and 1.73 seconds. Figure 2 shows the fitted curve alongside the data.

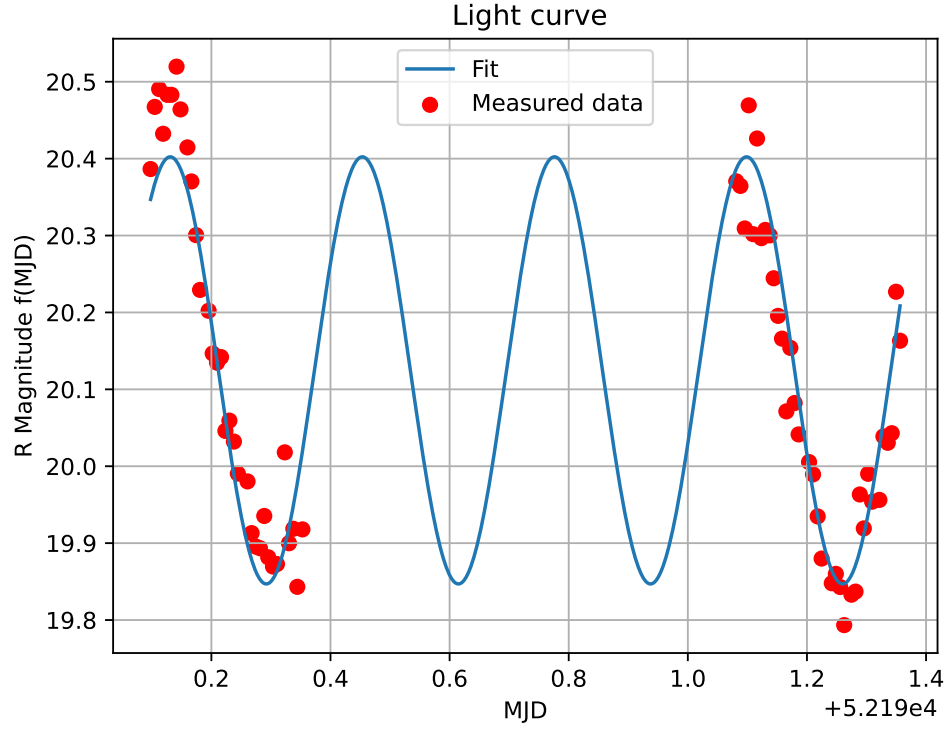


Figure 2: Sinusoidal fit to the light curve of 1999 TD10.

6 Conclusion

The photometric analysis of Kuiper Belt Object 29981 (1999 TD10) revealed a rotation period of approximately 15 hours, 29 minutes, and 1.73 seconds. The light curve analysis demonstrated periodic brightness variations, highlighting the object's rotational dynamics.